

REFLECT ON MODEL

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Er oppgaven båndlagt (konfidensiell)?	Ja / Nei
Er oppgaven unntatt offentlighet?	Ja / Nei

0.1 Tools

- ChatGPT: used for grammar editing and improving writing style
- Overleaf: used for writing and formatting the report

0.2 Purpose

0.2.1 What is the intended use of your model?

The purpose of the model is to investigate how different traffic compositions affect overall highway performance. Specifically, the model aims to:

1. Evaluate the potential advantages of fully human-driven, mixed, and fully autonomous traffic scenarios on highways.
2. Compare the performance of human-driven versus autonomous vehicles under identical conditions.
3. Focus on key factors such as traffic flow, reaction times, congestion, and road capacity in a two-lane highway setting.
4. Develop a simplified representation of vehicle-to-vehicle interactions, avoiding unnecessary complexity.

0.2.2 What is out of scope for this study

To keep the model focused on its main objectives, the following aspects are excluded:

1. Weather conditions such as rain, snow, and fog.
2. External disturbances including accidents, construction work, or vehicle breakdowns.
3. Collisions between vehicles. While collisions can cause congestion in reality, they are omitted to maintain focus on reaction times and inter-vehicle distances.
4. Roadside objects (trees, barriers, or infrastructure), as they are static and do not directly influence congestion, except in rare extreme events.
5. Unpredictable events involving humans or animals entering the road.
6. Mechanical failures. All vehicles are assumed to remain fully operational.
7. Traffic lights and other intersection controls, since the study focuses on uninterrupted highway driving.

0.3 Usage

0.3.1 What are typical usage situations for your model?

The model is designed to explore how different vehicle compositions affect highway traffic dynamics. Typical usage situations include:

1. Comparing traffic throughput and congestion patterns between scenarios with 100% human drivers, 100% autonomous vehicles, and mixed compositions (25%, 50%, 75% autonomous).

2. Analyzing optimal highway capacity utilization under different vehicle composition scenarios.
3. Adjusting parameters such as reaction time and following distance to observe their impact on traffic flow.
4. Conducting sensitivity analysis to determine which parameters (reaction time, following distance, vehicle mix ratio) have the greatest influence on traffic performance metrics.
5. Analyze how disturbances (braking, merging) propagate differently in human-driven versus autonomous vehicles connected car traffic.
6. Educational demonstrations showing how individual vehicle behaviors aggregate to system-level traffic patterns.

0.3.2 How and when do you (or the actors) interact with the model?

1. Vehicles: human-driven cars and Teslas interact with each other continuously during the simulation. Their interactions include braking, accelerating, merging, and maintaining safe distances.
2. Human drivers: introduce variability in reaction time and driving style, which the model uses to simulate unstable traffic conditions.
3. Initial Setup: Users configure simulation parameters including highway length, number of lanes, vehicle composition ratios, and individual vehicle characteristics (reaction times, following distances).
4. Parameter Adjustment: Users modify key variables between simulation runs to conduct comparative studies and sensitivity analyses.
5. Data Collection: At simulation completion, users extract quantifiable results including throughput measurements, average travel times, congestion formation patterns, and system efficiency metrics.
6. Scenario Comparison: Users run multiple simulation instances with different parameter sets to generate comparative datasets for analysis and reporting.

0.3.3 Scenarios. Describe 2-3 scenarios that you want to handle, how they relate to the purpose, and how they look in the usage.

Scenario 1: Phantom Jam Formation

Following Definition 2.4(execution as timed collection of snapshots)[2], this scenario illustrates one of the most common types of congestion: a phantom traffic jam. Phantom jams are unique because they do not require any external cause such as an accident or a bottleneck. Instead, they emerge purely from the dynamics of human driving, where delayed reactions and overcompensation turn a small disturbance into a full traffic wave.

1. **Initial State:** The system begins with 50 vehicles traveling at 100 km/h with a uniform 2-second spacing between them. Under these conditions, the road is in a free-flowing state, and no congestion is present. This reflects an ideal snapshot of smooth traffic flow.
2. **Disturbance:** At $t = 60s$, vehicle number 25 slightly reduces its speed by 10 km/h. This reduction might be caused by distraction, a misjudgment of spacing, or an overreaction to a perceived hazard. Although small in magnitude, this disturbance introduces instability into the system.
3. **Human-driven execution:** In the human-driven case, each following driver reacts with a delay, typically around one second. To compensate for the slower reaction, drivers tend to brake more harshly than the car in front. This overreaction amplifies as it propagates backward through the traffic stream. After only a few dozen vehicles, the ripple grows into a stop wave, forcing cars at the back of the platoon to a complete halt by $t = 180s$. The free-flowing state has collapsed into a phantom jam without any external cause.
4. **Tesla execution:** In the autonomous case, Tesla vehicles are connected via vehicle-to-vehicle (V2V) communication. When the disturbance is detected by vehicle 25, it immediately broadcasts the information to all Teslas behind it. Instead of delayed and isolated reactions, the following Teslas adjust their speeds smoothly and simultaneously. This synchronized response prevents the overreaction cycle from taking hold. By $t = 90s$, the flow is restored, and the system remains stable, avoiding the formation of a jam entirely.
5. **Purpose relation:** This scenario directly addresses the central research question of our project: whether connected autonomous vehicles can prevent congestion that arises solely from human behavior. In usage, the model demonstrates two contrasting executions: one where human drivers create instability and one where Teslas absorb the disturbance. This highlights the model's purpose of exploring if V2V communication can neutralize phantom jams, the most common form of congestion in real traffic systems.
6. **Usage:** Researchers configure sets of parameters and run comparative simulations to quantify the performance impact of introducing autonomous vehicles into human-driven traffic.

Scenario 2: Variation in Driver Behavior

1. **Initial State:** Mixed traffic flow with drivers exhibiting diverse styles. Some maintain large gaps and cautious speeds, while others follow closely and accelerate aggressively. Reaction times vary widely among the drivers.
2. **Disturbance:** An aggressive driver accelerates rapidly after a slowdown, while the following cautious driver reacts slowly, leaving a gap that later requires sudden braking.

3. **Human-driven execution:** The heterogeneous driving styles amplify small inconsistencies. Aggressive drivers create sharp accelerations, cautious drivers introduce delays, and the inconsistencies accumulate into stop-and-go waves. Even without external bottlenecks, congestion emerges due to behavioral differences.
4. **Tesla execution:** Connected Teslas maintain uniform, near-zero reaction times and smooth acceleration/braking patterns. When density increases, coordination prevents uneven spacing. Small fluctuations are absorbed rather than amplified, keeping the traffic stream stable and resistant to stop-and-go waves.
5. **Purpose relation:** This scenario examines if human variability itself can generate congestion. By standardizing reaction times and synchronizing vehicle behavior, connected Teslas can eliminate this instability and maintain steady flow.
6. **Usage:** Traffic engineers use this scenario to understand optimal autonomous vehicle distribution strategies for minimizing congestion spread.

Scenario 3: Merging at a Bottleneck

1. **Initial State:** A two-lane highway with steady traffic flow approaching an on-ramp. Vehicle density is near saturation, making efficient merging crucial.
2. **Disturbance:** Vehicles on the ramp attempt to merge but misjudge timing, while highway drivers hesitate to adjust speed. A single braking event triggers a chain reaction.
3. **Human-driven execution:** Uneven braking cascades through following vehicles, generating turbulence. Stop-and-go waves develop at the bottleneck, reducing throughput below road capacity and sustaining congestion for several minutes.
4. **Tesla execution:** Connected Teslas employ a “digital zipper merge.” On-ramp vehicles communicate speed and intent, while highway Teslas cooperatively adjust speed and spacing. The merging is completed smoothly without sudden braking, maintaining near-optimal flow and capacity.
5. **Purpose relation:** This scenario emphasizes bottlenecks as a primary real-world cause of congestion. By showing how connected vehicles coordinate to neutralize bottlenecks, the model demonstrates the value of V2V communication in sustaining free-flowing traffic.
6. **Usage:** This scenario can be used to quantify capacity improvements possible with autonomous vehicle adoption.

0.3.4 Data. What kind of data do you want to produce with the model, and how do you process the data? What type of data does the model need before it is ready for interactions?

For our model to function and produce meaningful results, it requires a well-defined set of input parameters and it must be designed to generate a specific output data for analysis. This section details the data required to configure the model and the data it produces to evaluate traffic system performance

Input Data

Before a simulation experiment can be run, the model requires a set of input parameters to define the initial state and behavioral rules of the system's objects. This initial data ensures that each simulation run is reproducible and grounded in realistic assumptions and we will start with a simple model of a straight, multi-lane highway stretch.

Input parameters:

1. **System Parameters:** These define the static environment in which the simulation takes place. This includes the length of the highway segment, the *number_of_lanes*, and the posted *speed_limit*
2. **Scenario Parameters:** These parameters configure the specifics of an experimental run, such as the initial *vehicle_density* (number of vehicles) on the highway and the *percentage_of_autonomous_vehicles* (e.g., the percentage of vehicles with *driver_type* = autonomous).
3. **Vehicle Parameters:** This data defines the attributes of the individual Vehicle objects in the simulation.
 - **Static Attributes:** These are physical constants for all vehicles, such as *vehicle_length* and *max_acceleration* / *max_deceleration* capabilities.
 - **Behavioral Attributes:** These parameters are crucial for modeling the differences between human and autonomous drivers. We will use an average or variable *reaction_time* and *following_distance* based on numbers from established traffic engineering literature.

For vehicles with *driver_type* = autonomous (Tesla Model Y), we will use parameters such as *reaction_time*, *default_following_distance* and a *communication_delay* to represent their ability to share data.

Output Data

Our goal is to start by generating simple data that shows the overall state of traffic flow. The model will primarily produce:

1. Time-series of vehicle states (position, speed, acceleration) with a resolution of 100 ms
2. Aggregated metrics: flow rate (vehicles/hour), average speed, density

This output data will be processed and visualized using simple plots. For example time-space diagrams to see how individual vehicle movements create larger patterns, like traffic waves.

0.3.5 Correctness. What constitutes the correctness of your model? How can you ensure that the model is correct?

Correctness is "the similarity of model executions with the referent system executions."^[2] We ensure this through:

Validation Data

A model's correctness is determined by the "similarity of the model executions with the referent system executions based on the model-of relationship and the perspective."^[2] To validate our model, we will compare its output to data from a reliable referent system. We plan to use two main sources for this comparison data:

1. **Real-World Data:** We will source publicly available traffic data from the Norwegian Public Roads Administration (Statens vegvesen). We will look for datasets containing aggregated traffic metrics, such as hourly flow rates and average speeds, for a relevant highway segment like the E18. We will then configure our model with similar road and traffic density parameters and compare our model's average outputs to these real-world measurements.
2. **Synthetic Data:** If matching specific real-world conditions proves too complex initially, we will first validate our model against a trusted, open-source traffic simulator like SUMO (Simulation of Urban MObility)^[1]. We can run a simple, human-only scenario in both our model and SUMO and compare the outputs. This ensures our baseline model behaves according to established traffic flow principles before we add the complexity of autonomous vehicles.

Bibliography

- [1] *Eclipse SUMO: Simulation of Urban MObility*. <https://eclipse.dev/sumo/>. Version 1.24.0 (accessed YYYY-MM-DD). Eclipse Foundation, 2025.
- [2] Andreas Prinz et al. *Understanding Modelling and Programming*. Springer, 2025. DOI: [10.1007/978-3-031-71280-7](https://doi.org/10.1007/978-3-031-71280-7).